Restoring the dynamic memory descendants
The second coefficient after a conserved subtraction

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Abstract

The previously projected second memory coefficient omits dynamic descendants of the lower fields. We restore them using the exact raw five-word identity and the established second-chaos table. An explicit polynomial commuting with the Hamiltonian removes the conserved terms. The remaining coefficient has a positive spectrum containing the quartic residual, the output-deficit residual, and energy-weighted lower fields. Its normalized small-deformation correlator is $(9\,\mathrm{sech}^8 - 2\,\mathrm{sech}^6)/7$, rather than the isolated projected sech^8 kernel.

1 State, scale, and the restored field

Use standard CAR, the full overlapping complex Gaussian p-body Hamiltonian, fixed charge $m = N/2$, normalized maximally mixed trace, and $\langle A, B\rangle = \mathbb{E}\,\mathrm{tr}_m\, A^\dagger B$. Set $\hbar = 1$, $h = H_m$, $\mathcal{J}_h A = \{h, A\}/2$, $\mathcal{L}_h A = [h, A]$, and

$$\lambda_N = p^2/N \to \lambda > 0, \quad q = e^{-4\lambda}, \quad C_N = \mathbb{E}\,\mathrm{tr}_m\, h^2 \to C = J_s^2/\lambda. \tag{1}$$

All chaos symbols refer to ordinary Gaussian coupling coordinates. The existing fields Y, F, R, T, Q_4 have diagonal two-energy measures [4]. The projected second coefficient was $X_N = (1-\widehat{\Pi}_N)mZ_2 = Q_{4,N} + 8\lambda_N C_N q_N^2 T_N + o_{L^2}(1)$. We now define an explicit conserved polynomial $\mathcal{C}_{2,N}$, given in Sec. 3, such that the actual remaining coefficient is

$$\boxed{U_{2,N} := mZ_2 - \mathcal{C}_{2,N} = X_N + b_R \mathcal{J}_h R + b_F \mathcal{J}_h F + b_J \mathcal{J}_h^2 Y + b_D \mathcal{L}_h^2 Y + C\kappa Y + o_{L^2}(1).} \tag{2}$$

Coefficients on the right can be evaluated at finite λ_N, q_N, C_N without changing the limit. This restoration subtracts only an operator commuting with h; the dynamic descendants have not been projected away. The constants are dimensionless, Y has energy dimension two, R, F have dimension three, and U_2 has dimension four.

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2 The exact raw identity behind the restoration

Let $A = \sum_i [H, c_i]^\dagger [H, c_i]$ restricted to m and let T_3 be the Hermitian third fermion-derivative insertion. On full Fock space define $DB = QB - \sum_i c_i^\dagger B c_i$ and $\mathcal{T}_j = (jp - D)H^j$. The five raw words are

$$E_0 = \{A, h^2\}/2, \qquad M_0 = hAh, \tag{3}$$

$$S_0 = \mathcal{J}_h \mathcal{T}_3 - E_0 - M_0, \qquad C_0 = \mathcal{T}_4 - 2E_0 - M_0 - 2S_0, \qquad Z_0 = h^4. \tag{4}$$

Restriction follows the full-Fock operations. Since $T_3 = \mathcal{T}_3 - 3\mathcal{J}_h A$, one has exactly

$$S_0 = \mathcal{J}_h T_3 + \mathcal{J}_h^2 A, \quad E_0 = \mathcal{J}_h^2 A + \mathcal{L}_h^2 A/4, \quad M_0 = \mathcal{J}_h^2 A - \mathcal{L}_h^2 A/4. \tag{5}$$

Set $d_3 = q^2 - q + 1$, $d_5 = 1 + q + q^2 + q^3 + q^4$. The old quartic projection coefficients are

$$\alpha = -\frac{2q^3}{d_3 d_5}, \qquad \beta = -\frac{q^2(1 + q + q^2)}{d_3 d_5}, \qquad \gamma = \frac{2q}{d_3}, \tag{6}$$

$$\delta = \frac{2\lambda(1 - q)^3(1 + q)(1 + q + q^2)}{(1 + q^2)d_5}. \tag{7}$$

Define $\mathcal{Q} = C_0 - \alpha E_0 - \beta M_0 - \gamma S_0 - \delta h^4$ and $Q_4 = \mathsf{W}_4 \mathcal{Q}$. For the raw memory core $\mathcal{E} = C_0 - 2S_0 + (E_0 + M_0)/2$, the exact identity is

$$\mathcal{E} = \mathcal{Q} + g_J \mathcal{J}_h^2 A + b_D \mathcal{L}_h^2 A + b_R \mathcal{J}_h T_3 + \delta h^4, \tag{8}$$

where $g_J = \alpha + \beta + \gamma - 1$, $b_D = (\alpha - \beta)/4$, and $b_R = \gamma - 2$. This is an identity of five raw words, before any asymptotic limit.

Let $B = h^2 - C_N I$, $O = \mathsf{W}_2 A$, and $e_{\mathcal{Q}} = \mathbb{E}\,\mathrm{tr}_m\, \mathcal{Q}$. The previously established lower-chaos table [3] gives

$$\mathcal{Q} = Q_4 + e_{\mathcal{Q}} I + C_N[p\ell_Q + k_Q]B + C_N o_Q O + j_Q \mathcal{J}_h F + 8\lambda_N C_N q_N^2 \mathcal{S}_N + o_{L^2}(1). \tag{9}$$

All coefficients in growing terms retain their finite-N parameters. In terms of the five table rows,

$$\ell_Q = 2q^2 - 2\alpha - 2\beta - 2q\gamma, \qquad o_Q = 1 - \alpha - q^2\beta - q\gamma = \frac{(1 - q)(1 + q)^2}{d_5}, \tag{10}$$

$$j_Q = 2q - \gamma, \qquad k_Q = k_C - \alpha k_E - \beta k_M - \gamma k_S - \delta k_Z, \tag{11}$$

with $k_E = -2\lambda q^2$, $k_M = -2\lambda(2q - 1)$, $k_S = -2\lambda(q^2 - q + 1)$, $k_C = -2\lambda(8\lambda q^2 - 3q^2 + 2q + 2)$, and $k_Z = q^2 + 2q + 3$.

3 An explicit commuting counterterm and controlled errors

Use the exact finite-sector definitions

$$a_N = \mathbb{E}\,\mathrm{tr}_m\, A, \qquad c_N = \frac{\langle O, B\rangle}{\langle B, B\rangle}, \qquad K_A = a_N I + c_N B, \qquad Y = A - K_A. \tag{12}$$

The earlier cubic subtraction is $T_3 - P_N = R + a\mathcal{J}_h Y + F - aB_S - bB_Z$, where $B_S, B_Z = O_{L^2}(p^{-1})$ and

$$a = -\frac{(1-q)^2}{1+q^2}, \qquad b = \frac{2\lambda(1-q)(1+q^3)}{(1+q^2)(1+q+q^2)}, \tag{13}$$

$$P_N = (ac_N + b)h^3 + [\bar{t} - (ac_N + b)\bar{z}]h. \tag{14}$$

Here $\bar{t}$ and $\bar{z}$ are the exact projections of $\mathsf{W}_1 T_3$ and $\mathsf{W}_1 h^3$ onto h in the annealed inner product. Let $r_N = p/m$ and define the already verified renewal counterterm

$$D_{2,N} = 2r_N h P_N + 3r_N^2 h^2 K_A - K_A^2/m - m r_N^4 h^4. \tag{15}$$

The exact renewal identity implies $mZ_2 = D_{2,N} + \mathcal{E} + O_{L^2}(p^{-1})$ at fixed λ. In particular the finite contribution from A^2/m remains in $D_{2,N}$.

Put $c = 2\lambda(1-q)/(1+q)$, $v = 2\lambda(1-q) + 16\lambda^2 q/(1+q)$, $d_v = 2\lambda(1+q)$, and $\rho = v/(v+d_v)$. The true finite-span projection gives $\mathcal{S}_N = cB + \rho Y + T_N + o_{L^2}(1)$. Now choose

$$\begin{aligned}\mathcal{C}_{2,N} = D_{2,N} &+ e_{\mathcal{Q}} I + \delta h^4 + g_J h^2 K_A + b_R h P_N \\ &+ C_N[p\ell_Q + k_Q + o_Q c_N + 8\lambda_N q_N^2 c]B.\end{aligned} \tag{16}$$

Every term commutes with h for each sample. The expectation $e_{\mathcal{Q}}$ is a finite Gaussian quartic contraction, not a fitted random scalar. Equation (16) keeps the exact $C_N, c_N, e_{\mathcal{Q}}$ and finite q_N, λ_N in all potentially growing subtractions.

Substituting into (8)–(9) proves (2), with

$$b_R = -\frac{2(1-q)^2}{d_3}, \qquad b_F = 2(q-1), \qquad b_D = \frac{q^2}{4d_5}, \tag{17}$$

$$b_J = \frac{(1-q)^2(q^4 - q^3 - 2q^2 - q + 1)}{(1+q^2)d_5}, \qquad \kappa = \frac{(1-q)(1+q)^2}{d_5} + 8\lambda q^2 \rho. \tag{18}$$

Here $b_J = g_J + a b_R$ and $b_F = b_R + j_Q$. The raw table has a proved $o_{L^2}(1)$ error. Multiplication of B_S, B_Z by h is controlled by the established factorial rank tails and Gaussian fourth-moment bounds. The renewal remainder uses fixed-degree marked overlap moments and Hölder. Thus the restoration uses actual strong remainders, not just formal chaos bookkeeping. It makes no claim of a simultaneous $\lambda \to 0$ limit.

4 Complete two-energy spectrum and the raw Taylor interface

Write $e = (E + E')/2$, $\omega = E - E'$, and

$$H_2(E, E') = b_J e^2 + b_D \omega^2 + C\kappa. \tag{19}$$

The diagonal basis Y, F, R, T, Q_4 and (2) give the positive measure

$$\boxed{d\nu_{U_2} = V_4 d\mu_{q^4} + V_2 d\mu_{q^2} + b_R^2 e^2 V_R d\mu_{q^3} + b_F^2 e^2 V_F d\mu_q + W H_2^2 d\mu_{q^2}.} \tag{20}$$

All μ_r are the existing normalized Mehler measures [1, 2]. The weights W, V_F, V_R, V_4, V_2 are the explicit positive weights of the preceding notes [4]; $W = 4J_s^4[\lambda^{-1} + 4q/(1+q)]$. The lower-field cross measures are now generally nonzero:

$$d\nu_{Y,U_2} = W H_2 d\mu_{q^2}, \tag{21}$$

$$d\nu_{L,U_2} = aeW H_2 d\mu_{q^2} + b_F e V_F d\mu_q + b_R e V_R d\mu_{q^3}. \tag{22}$$

For the common product energy marginal let K_j be the density of μ_{q^j}. Put $w = WK_2$, $f = V_F K_1$, $r = V_R K_3$, and $z = V_4 K_4 + V_2 K_2$. The full (Y, L, U_2) matrix factorizes as

$$M \operatorname{diag}(w, f, r, z) M^T, \qquad M = \begin{pmatrix} 1 & 0 & 0 & 0 \\ ae & 1 & 1 & 0 \\ H_2 & b_F e & b_R e & 1 \end{pmatrix}. \tag{23}$$

Its determinant is $w[(f + r)z + e^2 fr(b_F - b_R)^2] > 0$. All measures are absolutely continuous at fixed $0 < q < 1$, so the conserved-subtracted limiting field has no static atom. The L cross vanishes at ordinary center time by oddness under $(E, E') \mapsto (-E', -E)$; its energy-resolved measure must be retained.

Unlike the previous projected polynomial, the polynomial

$$\Psi_N(s) = Y_N + isL_N - s^2 U_{2,N}/2 \tag{24}$$

is exactly the second-order Taylor polynomial of $m\widehat{M}(s)$ after subtracting $\mathcal{C}_{0,N} + is\mathcal{C}_{1,N} - s^2\mathcal{C}_{2,N}/2$. The first two counterterms are those defined previously. At center time T its limiting covariance is

$$WG_{q^2}(T) + suC_L(T) - \frac{s^2 + u^2}{2} C_{Y,U_2}(T) + \frac{s^2 u^2}{4} C_{U_2}(T). \tag{25}$$

This identifies the actual Taylor coefficients. It does not yet bound the finite-memory remainder, nor determine other Taylor contributions such as the cubic coefficient paired with the linear one at the same total degree.

5 The ordered small-deformation limit changes the shape

After taking the fixed-λ double-scaled limit, let $\lambda \downarrow 0$ at fixed string time. The exact coefficients give

$$b_J = -16\lambda^2/5 + O(\lambda^3), \qquad b_R = -32\lambda^2 + O(\lambda^3), \tag{26}$$

$$b_F = -8\lambda + O(\lambda^2), \qquad b_D \longrightarrow 1/20, \qquad C\kappa \longrightarrow 16J_s^2/5. \tag{27}$$

The $C\kappa Y$ term is of leading size: κ starts at $16\lambda/5$, not at order λ^2. The energy-sum term is negligible in the relevant weighted norm, since e has moments of order $\lambda^{-1/2}$, while $b_J = O(\lambda^2)$. The R, F descendants and V_2 are also subleading after multiplying the total covariance by λ. Thus the two leading pieces are

$$Q_4, \qquad \frac{\mathcal{L}_h^2 + 64J_s^2}{20}\, Y. \tag{28}$$

Their leading norm squares, in units J_s^8/λ, are $768/5$ and $352/5$. The retained quartic residual carries $24/35$ of the total; the restored Y descendant carries $11/35$.

Let $f_j(T) = \operatorname{sech}^{2j}(2J_sT)$. The known probe limit and the uniform sixth frequency moment justify the fourth-derivative weighting of f_2. In particular

$$f_2''/J_s^2 = 64f_2 - 80f_3, \tag{29}$$

$$f_2^{(4)}/J_s^4 = 4096f_2 - 16640f_3 + 13440f_4. \tag{30}$$

Equation (20) then gives

$$\lambda C_{U_2}(T)/J_s^8 \longrightarrow \frac{768}{5} f_4 + \frac{f_2^{(4)}/J_s^4 - 128f_2''/J_s^2 + 4096f_2}{100}$$

$$= 288f_4 - 64f_3, \tag{31}$$

$$\boxed{\frac{C_{U_2}(T)}{C_{U_2}(0)} \longrightarrow \frac{9\operatorname{sech}^8(2J_sT) - 2\operatorname{sech}^6(2J_sT)}{7}.} \tag{32}$$

The corresponding norm and normalized curvature are

$$\lambda\|U_2\|_2^2 \longrightarrow 224J_s^8, \qquad \frac{\|[h, U_2]\|_2^2}{\|U_2\|_2^2} \longrightarrow \frac{240}{7} J_s^2. \tag{33}$$

Also $\lambda C_{Y,U_2}(T)/J_s^6 \to 16f_3(T)$. The sign change of (32) at late time is compatible with spectral positivity: the underlying measure is the positive sum in (20), with the Y contribution weighted by $(\omega^2 + 64J_s^2)^2/400$. Small λ is taken after the double-scaled limit; the estimates here are not uniform in N and λ jointly.

6 Executed checks and the remaining finite-memory task

Algebra and moments. The old raw contraction table was read directly from its saved certificate. A five-coordinate free-algebra check verifies (8); the simplified coefficients, all eight required unit-variance Mehler moments through degree six, and the Laurent coefficients are saved. For example, with probe parameter r,

$$\langle e^2\rangle_r = (1+r)/2, \quad \langle \omega^2\rangle_r = 2(1-r), \quad \langle e^2\omega^2\rangle_r = (1+q)(1-r^2)/2. \tag{34}$$

The leading norm pieces sum to 224 and curvature pieces to 7680; the latter divided by the former gives 240/7.

Independent finite CAR matrices and compressed-operator renewal give 16 exact identities across four nonzero systems, including $(N, p, m) = (6, 2, 3)$. They check the raw reconstruction, the renewal counterterm, and its commutation with h. Rational projection coefficients and arbitrary commuting K_A, P test the algebra; these checks do not assume finite-N equality to the limiting spectrum.

Weighted kernels. For $J_s = 1$, the actual normalized raw correlators are

λ	$T = 0.25$	$T = 0.5$	$T = 0.75$	$T = 1$
0.25	0.80095935	0.40558907	0.12737586	0.02987611
0.10	0.61782913	0.14910489	0.02044030	0.00477960
0.025	0.44492456	0.05075047	0.00420429	0.00082201
0.010	0.39448117	0.03253403	0.00153618	0.00023278

Jacobi cutoffs $38, 57, 146, 314$ have a weighted Taylor truncation bound below 10^{-12} for the normalized self kernel, with twelve extra levels for refinement. Independent 128/192-node Mehler integrals at $\lambda = 0.25, 0.1$ agree to 3.52×10^{-14}. They also verify the norm, curvature, and Y cross. The finite values at $T = 1$ remain positive even at $\lambda = 0.01$; they are not presented as already equal to the negative asymptotic tail. Absolute errors and phase-cancellation condition numbers are saved; floating and quadrature checks are not interval certification.

Continuation. The dynamic descendants are now restored, and the actual conserved-subtracted second coefficient is spectrally identified at half filling. The next tasks are a uniform central-charge extension and a two-insertion finite-memory remainder after an appropriate energy-dependent subtraction. The Taylor polynomial above supplies fixed coefficients but cannot justify exchanging the memory-time expansion with the double-scaled limit. Its connection to the Susskind meson spatial and mass dictionary remains a later reconstruction step.

References

[1] J. Tall and S. Tomsovic, arXiv:2604.14522v2, Sec. 4: strength density and energy-basis two-point functions.

[2] M. Berkooz, V. Narovlansky and H. Raj, arXiv:2006.13983. Existing complex-SYK chord and probe kernels.

[3] A. Day and Chengheng, local notes: *Restoring the raw memory descendants*; *Second memory coefficient and fourth singlet* (2026-11-06).

[4] A. Day and Chengheng, local notes: *Two orientations of the fermionic force*; *Mixed spectra of the first memory fields* (2026-11-06).